Time value of money

-4   -3   -2   -1   0   1   2   3   4   5

$\downarrow$

current

month/ year/ second/

hour

- 0 = P.V (Present value)
  5 = F.V (Future value)

- 0 → 5 (compound value will be used)
  5 → 0 (present value will be used)

- AIR - Adjusted inflation rate

- RRR - Required rate of return] Discounted
  RR - Rate of return

- RR
  ↳ K_e
  K_p
  K_d
  K_{RE}

- PVIF -
  CVIF - Compound value interest factor

- PViFA -

CViFA - Compound value interest factor annuity

- 3 important rates:

1. Inflation rate
2. Discounted rate
3. Interest rate

Q) Investor deposits ₹100 in a bank account for 5 years at 8% interest. Find out the amount which he will receive in his bank account if interest is compounded (i) annually (ii) semi-annually (iii) quarterly.

→ (i) ~~₹15~~ ₹ 146.93.

(ii) Semi annualy mein 8% ka half hoga as 6 months mein payout milega.

∴ 4% 10 baar milega as 5 years × twice a year

∴ ans → ₹148.02.

(iii) 2% as quarterly

= ₹ 148.59.

- Monthly:

→ 60 times (12 months × 5 years)

monthly = $\frac{8}{12}$ = 0.666666

= ₹148.98.

Q> If he invests Rs.100 every year

$I = 8\%$

$n = 5$ years

(i) Yearly

(ii) Semi anualy

(iii) Quarterly.

→

Year		Years of interest		
1	100	5	=	× 100
2	100	4	=	× 100
3	100	3	=	× 100
4	100	2	=	× 100
5	100	1	=	× 100

Lengthy method.

not to be done ×

= On calculator,

1.08 × (=) 4 times, then GT

= for semi annually,

1.04 × (=) 10 times, then GT, ~~and divide by~~ 2 multiply by 50

= for quarterly,

1.02 × (=) 20 times, then GT, multiply by 25

answers =

(a) 525.59
(b) 572.31
(c) 594.08

Q) If the discount / required rate is 10%, compute the present value of the cash flow streams detailed below :

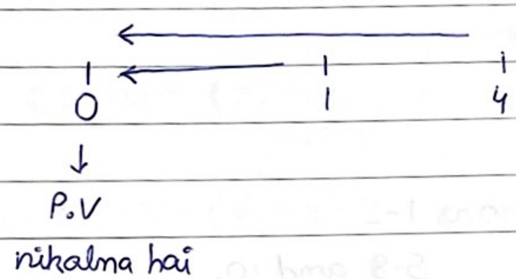
(i) Rs. 100 at the end of year 1

(ii) Rs. 100 at the end of year 4

(iii) Rs. 100 at the end of year 3 & 5

(iv) Rs. 100 for the next 10 years

(v) Rs.100 for the year 1 to year 5 and year 7 to year 10.



→ $1 \div 1.10$ (10%)

= 1 time

= ₹90.90 (at the end of year 1)

(ii) $1 \div 1.10$

= 4 times

= ₹68.30 (year 4)

(iii) $1 \div 10$

= 3 & 5 times

= 3 (75.13)

5 (62.09)

+ 137.22

(iv) $1 \div 10$

= 10 times, then GT (total)

= ₹ 614.45.

(v) 1 ÷ +θ 1.10

= 10 times (6th pe M+) (then charo GT)

= GT - (MRC)

= ₹558.00.

(vi) Fund for years 1-2

5-8 and 10.

= 1 ÷ 10

= 10 times (3, 4, 9 pe M+) (then GT)

= (GT - MRC)

= ₹428.61

Q> An executive is about to retire at the age of 60, His employer has offered 2 post retirement choices - 20 lakh rupees lumsum and ₹2,50,000 for 10 years. Assume 10% rate of interest. Which option is better?

-> Present value of ₹2,50,000 over 10 years

= 1 ÷ 1.10 = (10) times, then GT

= × 2,50,000 = ₹15,36,141.78

∴ 20 lakhs option is better

Formula =

Present value = $A\ (PVIFA)_{rn}$ $\rightarrow$ GT value

$= 2{,}50{,}000\ (PVIFA)_{0.10,0}$ [$1 \div 101.10$, then = 10 times then GT]

$= 2{,}50{,}000\ (6.1445)$

= ₹ 15,36,141.78

Note - The present value of option A is rupees 20 lakh while the present value of option B is ₹ 15,36,141.78. Hence, with the higher present value, option A is better.

Q> Compute the present value of perpetuity of ₹100 per annum if the discount rate is 10%.

-> Perpetuity = infinite years. / unlimited years

Formula = $PV = \frac{A}{I}$

$PV = \frac{100}{0.10}$

= ₹1000.

[Perpetuity]

Q> ABC limited has ₹10 crore bonds outstanding. Bank deposits earn 10% per annum. the bonds will be redemeed after 15 years for which purpose ABC limited wishes to create a sinking fund. How much amount should be deposited to the syning fund each year so that ABC limited would have in the sinking fund ₹10 crore to retire its entire issue of bonds.

→ Future value - 10 crore
Present value - ?

1.10 X = (15) times
= 4.177248
↓
GT
↓
33.84972

$$F.V = A\,(CVIFA_{rn})$$

$$10{,}00{,}00{,}000 = A\,(33.8497)$$

$$A = \frac{10{,}00{,}00{,}000}{33.8497}$$

$$= 0.0000033849$$

↓
Press ÷ =

= ₹29,54,297,02502

Q> ABC limited has borrowed ₹30 lakhs from CanBank Home Finance limited to finance the purchase of a house for 15 years. The rate of interest on such loans is 24% per annum. Compute the amount of annual payment/installment.

→ P.V = 30 lakhs
F.V = ?

= 1 ÷ 1.24 = (15)times

↓

GT

↓

÷ 30 lakhs

↓

÷ =

↓

₹ 7,49,760.08 / annually.

Q> An individual is having 2 alternatives – (i) ₹1,000 per annum for coming 10 years and (ii) ₹ 2,000 in year 2, 4, 6, 8, 10. Find out the better alternative by considering prevailing interest rate of 8.75%.

→ 6,488.88 (i) 6,838.58 (ii)

(ii) alternative is better

(i) $1 \div 1.0875 = (10 \text{ times})$

↓

GT

↓

×1000

$= \underline{6488.88}$

(ii) $1 \div 1.0875 = (10 \text{ times})$

↓

MR+ at 2,4,6,8,10 steps

↓

MRC

↓

×2200

$= \underline{6838.58}$.

Q> The earning of Fair growth Ltd. were ₹3 per share in year 1. They increased over a 10 year period to ₹4.02. Compute the rate of growth or compound annual rate of growth of the earnings per share.

$= \quad 4.02 = 3\,(CVIF)_{r,n}$

$4.02 = 3\,(CVIF)_{r,10}$

$\frac{4.02}{3} = CVIF_{r,10}$

$1.34 = CVIF_{r,10}$.

$r = 3\%$, $(1.03 \times = 10 \text{ times} = 1.34) \therefore \% = 3$

random method

$\therefore r = 3\%$

Q> Mr. X has deposit of ₹1,00,000 in a bank account for 3 years. (i) Assume it is compounded annually, semi annully and quarterly with stated interest rate of 4% per annum. Compute:

(a) (i) The amount he would have at the end of 3^{rd} year, leaving all interest paid on deposits in the bank

(b) Effective rate of interest he would on each alternative

(c) which plan should he choose?

Q> ABC Ltd. has borrowed ₹1,000 to be repayed in 12 monthly instalements of ₹94.56. Compute the annual interest rate.

Ans1.> (i) annually - ₹1,12,486.40 (4%) (1.04X = 2 times)

(ii) Semi annually - ₹1,12,616.24 (2%) (1.02 X = 5 times)

(iii) Quarterly - ₹1,12,682.50 (1%) (1.01x = 11 times)

(a) 1.04 X = 2 times

↓ = 4.16%

−1 → ÷3

(ii) 4.21%

(iii) 4.23% (Similar method)

(c) Quarterly is better

Ans2.) EMi = ₹ 94.56

n = 12 months

value = ₹1000.

= $\frac{1000}{94.56}$ = 10.575 or 10.58

$PV = A\,(PVIFA)_{r,n}$

$1{,}000 = 94.56\,(PVIFA)_{r,12}$

$10.58 = (PVIFA_{r,12})$

→ Now assuming the rate of interest:

if 12%, then each month 1%.

∴ $\frac{1}{1.01}$ = 12 times

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$15\% = \frac{1}{1.0125} = 11.08$ $(15\% \div 12)$

$18\% = \frac{1}{1.015} = 10.91$

$24\% = \frac{1}{1.02} = \underline{\underline{10.58}}$ (answer matched)

Q) ₹40,000 invested now grows to ₹70,000 in 5 years. Find the rate of return.

→ $\frac{70,000}{40,000} = \underline{\underline{1.75}}$.

- To find - rate of return.

= assuming $10\% = 1.10X = 4$ times $\times$ 40,000
= value not matched

↓

now $12\% = 1.12X = 4$ times $\times$ 40,000
= $\underline{\underline{₹70,493}}$.

∴ $\underline{\underline{12\%}}$ = answer

Q) ₹10,000 is deposited every year for 5 years. Interest rate is 9%. Find total value after 5 years.

= 1.09 x = 4 times
= 1.53862.

= 1.53862 × 10,000
= ₹ 15,386.23

Q) A bank agrees to receive ₹50,000 after 4 years for a loan. If the discount rate is 10%, find the present value of loan

→ 1 ÷ 1.10 = 4 times

0.6830 × 50,000 = ₹34,150 (PV)

Q) A firm saves ₹ 25,000 every year for 6 years to replace machinery. The interest rate is 10%. Find the total amount accumulated to replace machinery.

Q) A borrower agrees to pay ₹30,000 annually for 5 years to repay a loan. If the interest rate = 11%, find out value of loan.

Q) A project costs ₹2 lakhs, and yields ₹70,000 annually for 4 years. Discount rate is 10%. Should the project be accepted?

* very important formula -

$$K_e = \frac{D_1}{P_0} + g \quad , \quad \left[D_1 = D_0(1+g) \right]$$

$$\left[g = br \right]$$

D_1 = Dividend
K_e = cost of equity
g = growth
P_0 = Present value

r = return on investment
b = retention ratio

$$\therefore \left[P_0 = \frac{D_1}{(K_e - g)} \right]$$

↓

$$Eg: P_3 = \frac{D_4}{(K_e - g)}$$

(next year ka dividend liya jaayega)

$\therefore P_3$ mein D_4, P_6 mein D_7

Q) The required rate of return of investors is 14%. Assume the D_1 is ₹2.5. Compute the price at which the shares will be sold if the investors expect the earnings to grow by :

(i) 12% (ii) 14% (iii) 16%.

→ $r = K_e$

$K_e = 14\%$

$D_1 = ₹ 2.5$

(i) when $g = 12\%$

$$P_0 = \frac{D_1}{(K_e - g)}$$

$$= \frac{2.5}{0.14 - 0.12}$$

$$= \frac{2.5}{0.02}$$

$$= Rs.\ 125.$$

(ii) $g = 14\%$

$$P_0 = \frac{D_1}{K_e - g}$$

$$= \frac{2.5}{(0.14 - 0.14)}$$

$$= \frac{2.5}{0}$$

$$\therefore P_0 = \infty$$

(iii) $g = 16\%$

$$P_0 = \frac{D_1}{K_e - g}$$

$= \dfrac{2.5}{(0.14-0.16)}$

$= \dfrac{2.5}{(-0.02)}$

= undefined (negative value)

Q) Last year dividend was ₹8 per share, required rate of return is 15% and investors are expecting growth at 12%. Find present value of security.

→ $D_1 = D_0 (1+g)$

$= 8 (1+0.12)$

$= 8 (1.12)$

$= 8.96$

$P_0 = \dfrac{D_1}{(Ke-g)}$

$= \dfrac{8.96}{0.15-0.12}$

$= \dfrac{8.96}{0.03}$

$= ₹298.66$

- CAPM (Capital Assets Pricing Model):

$$\left[K_e = r_F + \beta\,(r_m - r_F)\right]$$

r_m = market return

r_F = risk free rate

β = unsystematic risk

[K_e can be replaced by r/r_e]

Q) risk free rate is 9%. Required rate of return is 18%. β of the share of ABC limited is 1.5. Expected dividend during the next year is 3. Growth rate in earnings at 8%. Find out value of share.

→ r_F = 9%
β = 1.5
r_M = 18%

$K_e = r_F + \beta\,(r_m - r_F)$

$= 0.09 + 1.5\,(0.18 - 0.09)$
$= 0.09 + 1.5\,(0.09)$
$= 0.09 + 0.135$
$= \cancel{0.0}\ 0.225$

$$P_0 = \frac{D_1}{(Ke-g)}$$

$$= \frac{3}{(0.225-0.08)}$$

$$= \frac{3}{0.145}$$

$$= ₹20.69$$

Q) Find out the return of securities by assuming the risk free rate of return is 6% and market return is 15%.

Security	β
A	1.5
B	0.9
C	2
D	0.5
E	5
F	0

→ $r_F = 6\%$

$r_M = 15\%$

(i) A :

$$Ke = r_F + \beta(r_m - r_F)$$

$$= 0.06 + 1.5(0.15-0.06)$$

$$= 0.06 + 1.5(0.09)$$

$$= 0.06 + 0.135$$

$= \cancel{0.0}\ 0.195\ /\ 19.5\%$

(ii) B

$$Ke = 0.06 + 0.9(0.15 - 0.06)$$
$$= 0.06 + 0.9(0.09)$$
$$= 0.06 + 0.081$$
$$= 0.141\ /\ 14.1\%$$

(iii) C

$$Ke = 0.06 + 2(0.09)$$
$$= 0.06 + 0.18$$
$$= 0.24\ /\ 24\%$$

(iv) D

$$Ke = 0.06 + 0.5(0.09)$$
$$= 0.06 + 0.045$$
$$= 0.105\ /\ 10.5\%$$

(v) E

$$Ke = 0.06 + 5(0.09)$$
$$= 0.06 + 0.45$$
$$= 0.51\ /\ 51\%$$

(vi) F

$Ke = 0.06 + 0(0.09)$
$= 0.06 + 0$
$= 0.06$ / 6%

Q) The recquired rate of return of a particular share is 15%. Last year dividend was ₹ 5 per share. It is expected to grow as below:

Year	g
1	8%
2	10%
3	7%
4	12%
5	6%

The value expected at end of 5th year is ₹ 200 per share. Find out present value.

→ $D_1 = D_0 (1+g)$
$= 5(1+0.08)$
$= 5 \times 1.08$
$= 5.4$.

$$P_0 = \frac{D_1}{(Ke-g)} = \frac{5.4}{(0.15-0.08)} = \frac{5.4}{0.07} = 77.142.$$

5.4

→ $D_2 = \cancel{5}(1+0.10)$
$= 5 \times 1.\cancel{10}$
$= 5.4 \times 1.10$
$= \underline{\underline{5.94}}$.

$D_3 = D_2 (1+g)$
$= 5.94 (1+0.07)$
$= 5.94 (1.07)$
$= \underline{\underline{6.36}}$.

$D_4 = D_3 (1+g)$
$= 6.36 (1+0.12)$
$= 6.36 (1.12)$
$= \underline{\underline{7.12}}$.

$D_5 = D_4 (1+g)$
$= 7.12 (1+0.06)$
$= 7.12 (1.06)$
$= \underline{\underline{7.55}}$.

Q) The recquirid rate of return on a particular security by an investor is at 12%. The expected dividend is ₹5 per share which will grow by 10% in year 2 and year 3, from year 4, the growth rate is expected to be 8% forever. Find out the present value of the security.

→ $D_1 = ₹5$
D.S.R = 12%.

$D_2 = 5.5$ (10% growth)

$D_3 = 6.05$ (10% growth)

$D_4 = 6.53$ (8%)

$$P_3 = \frac{D_4}{Ke - g}$$

$$= \frac{6.53}{(0.12 - 0.08)}$$

$$= \frac{6.53}{0.04}$$

$= ₹163.25.$

$D_3 + P_3 = 6.05 + 163.25$

$= ₹169.30.$

[when total value after years is not given, we need to find out]

[jiss year se constant hota hai growth, waha se find out karna hota hai].

Q) The required rate of return on a security is 16%, dividend received was ₹ 12 per share which is expected to grow in the first 3 years at 12%, and in year 4 and 5 by 10% and from 6th year onwards, it will be grown by 8% ~~10%~~. Find out present value of share.

→ $K_e = 16\%$

$D_0 = ₹12$

$D_6 = ~~20.~~22.03.$] 8%

$D_1 = 13.44$]

$D_2 = 15.05$] 12%

$D_3 = 16.86$]

$D_4 = 18.545$] 10%

$D_5 = 20.40.$]

$$P_5 = \frac{D_6}{(Ke-g)}$$

$$= \frac{22.03}{(0.16-0.08)}$$

$$= \frac{22.03}{0.08}$$

$$= ₹275.38.$$

$$(D_5 + P_5) = 20.40 + 275.38$$
$$= 295.78.$$

Present value = ₹184.64.

Yield to maturity

Q) A bond has 3 years remaining until maturity. It has a par value of Rs. 1000. The coupon rate of interest is 10%. Compute the yield to maturity at the current market price of (i) 1100 (ii) 1000 (iii) 900. Assume interest is paid annually.

→ R.V
net proceeds = ₹1000

I = 10% = ₹100

$n = 3$ years

np

(i) ~~M.V~~ = 1100

$$= \left[\frac{I + \left(\frac{R.V - np}{n}\right)}{\frac{RV + np}{2}} \right]$$

$$= \frac{100 + \left(\frac{1000 - 1100}{3}\right)}{\frac{1000 + 1100}{2}}$$

$$= \frac{100 + (-33.33)}{1050}$$

$$= \frac{66.67}{1050}$$

$$= \cancel{14} \; 6.349\%$$

(ii) $np = 1000$

$$= \frac{100 + \left(\frac{1000 - 1000}{3}\right)}{\frac{1000 + 1000}{2}}$$

$$= \frac{100 + (0)}{1000}$$

$$= 10\%$$

(iii) $np = 900$

$$= \frac{100 + \left(\frac{1000 - 900}{3}\right)}{\frac{1000 + 900}{2}}$$

$$= \frac{100 + 33.33}{950}$$

$$= \frac{133.33}{950}$$

$$= 14.03\%$$

Q) A preference share is having 5 years to maturity. The par value of a preference share is ₹250 per share. The coupon rate of dividend is 12%. Company follows different alternatives for this security.

(i) It can issue with 10% of discount and redeem at par.

(ii) It can issue at par and redemed at 10% premium.

(iii) It can issue 5% discount and redeem at 5% premium.

Find out YTM.

→ I = Preference dividend

(i) $np = 225 \quad RV = 250$

(ii) $np = 250 \quad RV = 275$

(iii) $np = 237.5 \quad RV = 262.5$

(i)
$$\frac{P.D + \left(\frac{RV - np}{n}\right)}{\frac{RV + np}{2}}$$

$$= \frac{30 + \left(\frac{250-225}{5}\right)}{\frac{250+225}{2}}$$

$$= \frac{30 + \left(\frac{25}{5}\right)}{237.5}$$

$$= \frac{35}{237.5}$$

$$= 14.74\%$$

(ii) $$\frac{30 + \left(\frac{275 - 250}{5}\right)}{\frac{275 + 250}{2}}$$

$$= \frac{30 + \left(\frac{25}{5}\right)}{262.5}$$

$$= \frac{35}{262.5}$$

$$= 13.33\%$$

(iii) $$\frac{30 + \left(\frac{262.5 - 237.5}{5}\right)}{\frac{262.5 + 237.5}{2}}$$

$$= \frac{30 + \left(\frac{25}{5}\right)}{\frac{500}{2}}$$

$$= \frac{35}{250}$$

$$= 14\%$$

Q) The shares of ABC Limited are currently selling for ₹100 on which the expected dividend is ₹4. Compute the total return on shares if earnings or dividends are likely to grow at (i) 5% (ii) 10% (iii) 0%. (K_e)

Q) ABC Limited paid a dividend of ₹2 per share last year, which is expected to grow at 10%. If the current market price is ₹40, and the required rate of return is 18%, compute the expected dividend yield and capital gains yield next year.

Q) The required rate of return of investors is 12%. D_0 = ₹5 per share. It is expected to grow at 20% in coming 3 years and at 10%, thereafter. Compute the current value of share.

Answers.

1.) $Ke = \frac{D_1}{P_0} + g$ (i)

$= \frac{4}{100} + 0.05$

$= \frac{1}{25} + 0.05$

$= 0.04 + 0.05$

$= 0.09\% / 9\%$

(ii) $= \frac{4}{100} + 0.10$

$= 0.04 + 0.10$

$= 0.14\%$ or 14%

(iii) $= \frac{4}{100} + 0\%$

$= 0.04$ or 4%

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2.) $D_0 = 2$, $D_1 = 2.2$ (10%)

$g = 10\%$, $P_0 = 40$

$K_e = 18\%$

Dividend yield $\cancel{K_e} = \frac{D_1}{P_0} \cancel{+ g}$

$= \frac{2.2}{40} \cancel{+ 0.1}$

$= 0.055 \cancel{+ 0.10}$

$= \cancel{0.155 \text{ or } 15.5\%}$

$= 5.5\%$ ✓

$P_1 = \frac{D_2}{K_e - g}$ → $D_1(1+g) = 2.42$

$= \frac{2.42}{0.08}$

$P_1 = 30.25$

$\therefore \frac{P_1 - P_0}{P_0}$ (To find capital gain)

$= \frac{30.25 - 40}{40} = (-24.38\%)$

$\therefore$ capital loss $= 24.38\%$

3.) $K_e = 12\%$

$D_0 = ₹5$

$D_1 = ₹6$

$D_2 = ₹7.2$

$D_3 = ₹8.64 + P_3 = ₹475.2$.

$D_4 = ₹9.504$

$P_3 = \frac{D_4}{(K_e - g)}$

$= \frac{9.504}{(0.12 - 0.10)}$

$= \frac{9.504}{0.02}$

$= 475.2$

Process $= \frac{1}{1.12} = \times 6 (D_1) = M+$

↓

same process and then MRC.

(= 2 times, = 3 times)

Final answer $= P_0 = 355.49$.

Q) Assume (i) ₹100 par value

(ii) 8% coupon rate of interest

(iii) 10 years remaining to maturity

(a) If interest is paid annually, find out the value of bond if the required rate of return is 7%, 8%, and 10%. Indicate for each of the case whether the bond is selling at a discount, at a premium or at its par value.

→ (when not given, assume par value = redeemed value)
∴ maturity at par value.

(i) at 7% = $P_0 = I_{1-10}(PVIFA_{r,n}) + P_{10th}(PVIF_{r,n})$

= ~~79.26 (7)~~ + 0.5083 (1 ÷ 1.07 (= 10 times))

Interest

= ~~7 (8)~~

↓

GT × 8 → M+

↓

0.5083 × 100 → M+

MRC = 107.01

(ii) at 8%

0.4631 (1 ÷ 1.08 (= 10 times))

↓

GT × 8 → M+

↓

0.4631 × 100 → M+

↓

MRC → 99.9906 ≃ 100

(iii) at 10%

0.3855

↓

GT × 8 → Mt

↓

0.3855 × 100 → Mt

↓

MRC.

₹ 87.71.

P_0 at 7% = 107.02 (Premium)
P_0 at 8% = 100 (Par)
P_0 at 10% = 87.71 (Discount)

$$\begin{bmatrix} K_e > r \\ K_e = r \\ K_e < r \end{bmatrix}$$

[8% Ke tha, jab interest rate 8% tha, toh par value aayi]

Q) A company has EPS of ₹20. It retains 40% of earnings Return on investment = 15%, cost of equity = 12%. Find out market value of share.

→ $D_0 = 8$ ₹ $g = \beta r$
(Dividend)

Q) A share is selling at ₹180. expected dividend next year is ₹14 with growth of 6%. Required rate of return is 11 should the share be bought?

→ Find P_0 and compare

Q) A bond of ₹1000 pays 10% interest anually for 3 years, then 12% for remaining 4 years. Required rate of return is 11%. Find the value of bond today.

→ Find P_0, Total years = 7.

Q) A zero coupon bond of ₹1000 matures after 8 years. The required rate of return is 9%. Find out its current value.

→ 1000 × PVIF 8 years at 9%

∴ 1 ÷ 1.09 (= 8 times)

↓

×1000

=

₹501.866

Present value

$$K_e = \frac{D_1}{P_0} + g$$
(equity)

$$K_d = I(1-t) \;\Big/\; \frac{I(1-t) + \left(\frac{R.V - \overset{N}{\cancel{M}}P}{n}\right)}{\frac{R.V + \overset{N}{\cancel{M}}P}{2}}$$
(debentures/debt)

preference dividend

$$K_p = \frac{PD}{\underset{N}{\cancel{M}}P} \;\Big/\; \frac{PD + \left(\frac{RV - \overset{N}{\cancel{M}}P}{n}\right)}{\frac{RV + \overset{N}{\cancel{M}}P}{2}}$$
(preference shares)

NP = net proceeds.

Q) 16% preference shares with the face value of Rs 350 per share was issued at 5% discount and it will be redeemed after 12 years at 8% premium. Company has paid Rs 2 per share as a floating cost and a registration charge at 0.5% on face value to exchange. Calculate the cost of a particular share. The company is falling under the tax bracket of 35%. [Dokha]

→ F.V = 350

Issue price = ₹ 332.5

n = 12 years

R.V = ₹ 378

Floating cost = ₹2

Registration charrge = ₹1.75

Tax = 35%

PD = Rs 56 (16%)

To find = K_p

$$K_p = \frac{PD + \left(\frac{RV - NP}{n}\right)}{\frac{RV + NP}{n}}$$

NP = ₹ 328.75

(332.5 - 2 - 1.75)

$$\therefore \frac{56 + \left(\frac{378 - 328.75}{12}\right)}{\frac{378 + 328.75}{2}}$$

$$= \frac{56 + \left(\frac{49.25}{12}\right)}{353.38}$$

$$= \frac{56 + 4.10}{353.38}$$

$$= \frac{60.10}{353.38}$$

= 0.1701

= 17.01%

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Date ______
Page ______

Q) 18% irredeemable preference shares with a face value of Rs 400 was issued at 2% discount. SEBI charges 3% registration cost and Rs. 4 as floating cost per share. Calculate the cost of preference share.

Q) 9.5% preference shares with a face value of Rs. 500 was issued at par and redeemable after 15 years at 20% premium. Applicable tax rate is 10% and underwriters charges are paid at 5% on issue price along with stamp duty on particular security was paid at 2%. Find out cost of security.

Q) 15% debt note with was issued with a face value of Rs125. with 5% premium. Having majority value at par after 10 years company needs to pay floating charges at 2%. underwriters are charging 3% on issue price. Applicable tax rate is 30%. Find out cost of debt note.

→ I^T applicable hoga and replace PD with I

Q) Dividend paid last year was Rs. 7 per share. The expected growth rate on a particular dividend is 8%. Current value of a share is Rs. 140. 12% preference shares was issued at 10% premium and at end, it will be redeemed at 25% premium, face value of this share is ₹300 and remained unpaid till 8 years. 18% debt note was issued at par and it will be redeemed after 10 years with 10% premium. Assume face value is ₹400. Applicable tax rate is 35%. Find out cost of each source of capital.

→ To find Ke, Kp, Kd :

= $D_0 = 7$

$D_1 = 7.56$ (8%)

→ $$\left[K_e = \frac{D_1}{P_0} + g \right]$$

$$= \frac{7.56}{140} + 0.08$$

$$= 0.134$$

$$= 13.4\%$$

$$\left[K_p = \frac{PD + \left(\frac{RV - NP}{n}\right)}{\left(\frac{RV + NP}{2}\right)} \right]$$ for redeemable

PD = 300×12%, RV = 300+25%, NP = 300+10%

$$\therefore \frac{36 + \left(\frac{375 - 330}{8}\right)}{\frac{375+330}{2}}$$

$$= \frac{36 + 5.625}{352.5}$$

$$= \frac{41.625}{352.5}$$

$$= 0.11808$$

$$= 11.81\%$$

$$\left[K_D = \frac{I(1-t) + \left(\frac{RV-NP}{n}\right)}{\frac{RV+NP}{2}} \right]$$ for redeemable debentures/ debt notes

$I = 72$ (18% of 400)

$RV = 440$ (400+10%)

$NP = 400$

$n = 10.$

$$= \frac{72(1-0.35) + \left(\frac{440-400}{10}\right)}{\frac{440+400}{2}}$$

$$= \frac{72(0.65)+4}{420}$$

$$= \frac{46.8+4}{420}$$

$$= \frac{50.8}{420}$$

$= 0.1209$

$= \underline{12.09\%}$

$$\left[K_P = \frac{PD}{NP}\right], \left[K_D = \frac{I(1-t)}{NP}\right]$$ for non-redeemable.

For loan = $\underline{I(1-t)}$.

- K_o (overall cost of capital):

→ Format:

Equity   Weight × Cost = weighted cost

Q) Company follows following capital structure:

Source of capital	Amount (₹)
Equity share	10,00,000
15% preference shares	5,00,000
18% preference shares	7,50,000
12% debentures	7,50,000
18% debt note	10,00,000

Additional info:

1. Expected dividend on a particular share is ₹ 8.5 which is expected to grow at 12% per annum considering current market price is ₹ 160. and face value was ₹ 100. per share

2. 15% preference shares with face value of ₹ 250 was issued at par and it will be redeemed after 8 years. with 8% premium the total cost of issuing this category of the share was 4%.

3. The current price of 18% of preference share is ₹96 which was issued with the face value of Rs 100.

4. 12% debentures was issued with the face value of Rs 50 per debenture at 10% discount and it will be redeemed at 15% premium after 12 years. Total cost of issuing this debenture is 3.5% on issue price (which includes floating cost, underwriter commision, cost of exchange etc).

5. 18% debt note was issued with a face value of Rs 75 per debt note and having current market price as ₹110 per debt note. ~~Pry~~ Prevailing tax rate is 40%. Calculate overall cost of capital.

→ $K_e = \frac{D_1}{P_0} + g$

$D_1 = 8.5$

$g = 12\%$

$P_0 = 160$

$K_e = \frac{8.5}{160} + 0.12$

$= 0.053125 + 0.12$

$= 0.173125$

$= 17.31\%$

K_p (redeemable) $= \dfrac{PD + \left(\frac{RV - NP}{n}\right)}{\left(\frac{RV + NP}{2}\right)}$

Dividend or Interest amount will be calculated on face value.

$PD = 37.5$
$RV = 270$
$NP = 2\cancel{5}\overset{}{4}0 \ (250 - 4\%)$
$n = 8$

$$K_p = \frac{37.5 + \left(\frac{270 - 2\cancel{5}40}{8}\right)}{\left(\frac{270 + 2\cancel{5}40}{2}\right)}$$

$$= \frac{37.5 + \cancel{2.5}\ 3.75}{\cancel{260}\ 255}$$

$$= \frac{\cancel{40}}{\cancel{260}} \quad \frac{41.25}{255}$$

$$= 16.17\%$$

3.) $K_p = \frac{PD}{NP}$

$\cancel{PD} = \cancel{96}$
$PD = 18 \ (18\% \text{ of } 100)$
$NP = 100$

$$K_p = \frac{18}{\cancel{100}\ 96}$$

$$= \cancel{18\%} = 18.75\%$$

4.) $$K_D = \frac{I(1-t) + \left(\frac{RV - NP}{n}\right)}{\left(\frac{RV + nP}{2}\right)}$$

$n = 12$ years

$RV = 57.5$

$NP = 45 - 3.5\%$

$= 43.425$

$t = 40\%$

$I = 12\%$ of 50

$= 6$

$$K_d = \frac{I(1-t) + \left(\frac{RV-NP}{n}\right)}{\frac{RV+NP}{2}}$$

$$= \frac{6(1-0.40) + \left(\frac{57.5-43.425}{12}\right)}{\frac{57.5+43.425}{2}}$$

$$= \frac{3.6 + \left(\frac{14.075}{12}\right)}{50.4625}$$

$$= \frac{3.6 + 1.173}{50.4625}$$

$$= 9.45\%$$

5.) Tax $= 40\%$

$I = 18\%$ of 75

$= 13.5$

$$K_d = \frac{I(1-t)}{NP}$$

$$= \frac{13.5(1-0.40)}{110}$$

$$= \frac{13.5 \times 0.60}{110}$$

$$= \frac{8.1}{110}$$

$$= 7.36\%$$

→

Source	Amount	Weight	Cost	W.C
Eq.	10,00,000	0.2500	0.1731	0.043275
15% pref	5,00,000	0.1250	0.1618	0.020225
18% pref	7,50,000	0.1875	0.1875	0.035156
12% deb.	7,50,000	0.1875	0.0946	0.017737
18%	10,00,000	0.2500	0.0736	0.0184
				0.1347

Q) After tax cost of capital of XYZ company limited is as follow

cost of debt = 8%
cost of preference share (including dividend tax) = 14%
cost of equity = 17%

Source	Book value amount (₹)	market value amount(
Debt	3,00,000	2,70,000
Preference share	2,00,000	2,30,000
Equity	5,00,000	7,50,000

Calculate WACC as per ber book book value weights and marke value weights.

Q) A company is considering raising Rs1,00,00,000 by one of the two alternative methods as follows:

1. 14% institutional term loan and
2. 13% non convertible debentures

The term loan option would attract no major incidental cost. The debentures would have to be issued at 2.5% discount and would also involve ₹1 lakh as a cost of issue. Assume a tax rate of 35%. Advice the firm which alternative should be accepted.

→ (i) Loan:

$14\% - 35\% = 9.1\%$

(ii) Debentures:

$13,00,000 - 35\% \times \div 96,50,000$

$= 8.76\%$

(i) Term loan

$I(1-t) = 0.14(1-0.35)$

$= 0.14 \times 0.65$

$= 0.091$

$= 9.1\%$

(ii) $\dfrac{I(1-t)}{NP} = \dfrac{13,00,000(1-0.35)}{96,50,000}$

$= 0.0876$

$= 8.76\%$

$NP = 1,00,00,000 - \underset{\text{(Discount)}}{2,50,000} - \underset{\text{(cost)}}{1,00,000}$

$= 96,50,000$

For [Poora paisa chahiye → Loan ✓
Cost kam chahiye → Debenture ✓]

Q) XYZ company is planning to issue 13% preference shares of ₹100 each redeemable at 8% premium after 8 years. They are expected to be sold at 2% premium. The likely floatation cost is ₹2 per share. Determine the cost of preference shares assuming dividend payment tax of 17% and income tax at 40%.

→ For dividend payment tax:

$$\frac{PD(1+D_t) + \left(\frac{RV-NP}{n}\right)}{\left(\frac{RV+NP}{2}\right)}$$

PD = 13 (13% of 100)
D_t = 17% (0.17)
RV = 108 (100+8%)
NP = 100+2% (premium) − 2% (floatation)
= ₹100
n = 8 years

$$= \frac{13(1+0.17) + \left(\frac{108-100}{8}\right)}{\frac{108+100}{2}}$$

$$= \frac{13(1.17)+1}{104}$$

$$= \frac{15.21+1}{104} = \frac{16.21}{104} = 15.58\%$$

Q) Assuming corporate tax at 35%, compute after tax cost of capital.

1. A perpectual 15% debentures of ₹1000 sold at premium of 10% with no floatation cost.

2. A 10 years 14% debentures of ₹2,000, redeemable at par, with 5% floatation cost.

3. A 10 year 14% preference share of ₹100. Redeemable at premium of 5% with 5% floatation cost.

4. An equity share selling at ₹50 and paying a dividend of ₹6 per share, which is expected to be continued indefinitely.

5. The same equity share, if dividend is expected to grow at 5%.

6. An equity share selling at ₹120 per share, of a company engages only in equity financing. The EPS is ₹20, out of which 50% is paid in dividends. The shareholders ~~Out~~ expect the companies to earn a constant after tax rate of return at 10% on its retained earnings.

Q) Considering the following figures, calculate the cost of equity for a particular security. You are also requested to advise which of the securities to be issued.

Security	R_F	R_M	β
A	6%	14%	0.08
B	5%	3%	1.10
C	7%	21%	2.9
D	8%	26%	2.5
E	9%	3%	0.7
F	7%	11%	0.4

→ 1. $K_e = r_F + \beta(r_m - r_f)$

$= K_e = 0.06 + 0.08(0.14 - 0.06)$

$= 0.06 + 0.08(0.08)$

$= 0.06 + 0.0064$

$= 0.0664.$

2. $K_e = r_f + \beta(r_m - r_f)$

$= K_e = 0.05 + 1.10(0.03 - 0.05)$

$= 0.05 + 1.10(-0.02)$

$= 0.05 - 0.022$

$= 0.028.$

3. $K_e = r_f + \beta(r_m - r_f)$

$= 0.07 + 2.9\ (0.21 - 0.07)$
$= 0.07 + 2.9\ (0.14)$
$= 0.07 + 0.406$
$= \underline{0.476}.$

4. $K_e = r_f + \beta\ (r_m - r_f)$

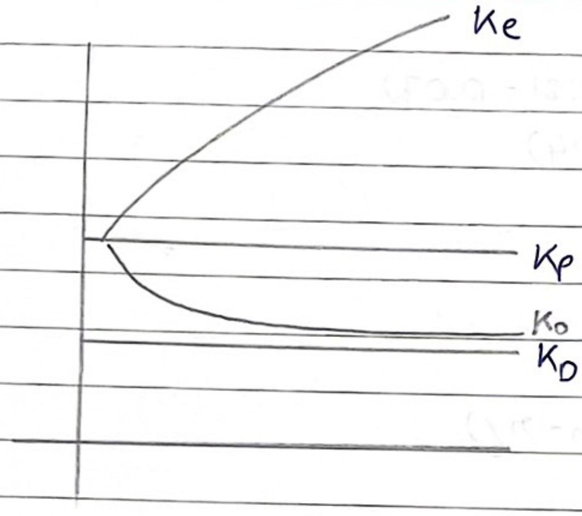
$= 0.08 + 2.5\ (0.26 - 0.08)$
$= 0.08 + 2.5\ (0.18)$
$= 0.08 + 0.45$
$= \underline{0.53}.$

5. $K_e = r_f + \beta\ (r_m - r_f)$

$= 0.09 + 0.7\ (0.03 - 0.09)$
$= 0.09 + 0.7\ (-0.06)$
$= 0.09 + (-0.042)$
$= 0.09 - 0.042$
$= \underline{0.048}.$

6. $K_e = r_f + \beta\ (r_m - r_f)$

$= 0.07 + 0.4\ (0.11 - 0.07)$
$= 0.07 + 0.4\ (0.04)$
$= 0.07 + 0.016$
$= \underline{0.086}.$



$K_d < K_p < K_e$
↓
highest

If company has all debt, $K_o = K_d$
If company is debt free, $K_o = K_e$.

Q) A company has on its books, the following amounts and specific cost of each type of capital:

Type of capital	Book value	Market value	Specific cost (%)
Debt	4,00,000	3,80,000	5
Preference	1,00,000	1,10,000	8
Equity	6,00,000	12,00,000	15
Retained earnings	2,00,000		13

Calculate WACC as per BV weights and MV weights.

$$\left[K_e = K_{RE} \right]$$ ☆

$\left[\text{where, } K_{RE} \text{ has not been given, assume } K_e = K_{RE}\right.$
$\left.\text{If } K_{RE} \text{ has been given, do not assume}\right]$

So, for this sum, $K_e \neq K_{RE}$, as both have been given.

= Combined market value of E and RE = 12 lakhs.

Total BV of both = 8 lakhs

8 → 12

6   2

∴ 6 → 9 ∴ MV of E = 9 lakhs
2 → 3 RE = 3 lakhs.

Book Value.

Type	Amount	weight	COC	WC
Debt	4,00,000	0.308	~~1.54~~ 0.05	0.0154
Preference	1,00,000	0.077	0.08	0.00616
Equity	6,00,000	0.4621	0.15	0.06915
RE	2,00,000	0.154	0.13	0.02002
	13,00,000			0.11073

↓

≃ 11.07%

Market value

Type	Amount	Weight	Cost	WC
Debt	3,80,000	0.225	0.05	0.01125
Preference	1,10,000	0.065	0.08	0.0052
Equity	9,00,000	0.532	0.15	0.0798
RE	3,00,000	0.178	0.13	0.02314
	16,90,000			0.11939

↓

≃ 11.94%

Q) You are required to calculate WACC which will be relevant for evaluating long term investment projects.

Cost of equity - 12%
After tax cost of long term debt - 7%
After tax cost of short term debt - 4%

Source	Book value	Market value
Equity	5,00,000	7,50,000
Long term debt	4,00,000	3,75,000
☆ Short term debt	1,00,000	1,00000
	10,00,000	12,25,000.

→ Pehle weight [Short term calculate nahi hoga iss sum mein

Book Value

Source	Amount	Weight	Cost	~~WACC~~ WC
Equity	5,00,000	0.555	0.12	0.0666
Long term Debt	4,00,000	0.445	0.07	0.03115
	9,00,000			0.09775

↓

≃ 9.78%

Market Value

Source	Amount	Weight	Cost	WC
Equity	7,50,000	0.667	0.12	0.08004
Long term Debt	3,75,000	0.333	0.07	0.02331
	11,25,000			0.10335

↓

≃ 10.33%

Q) ABC limited wishes to raise additional finance of ₹10 lakhs for meeting its investment plans. It has ₹2,10,[illegible] in the form of retained earnings available for investment purposes. Following are the details:

1. Debt equity mix is 30 : 70.

2. Cost of Debt upto ₹1,80,000, 10% (before tax), beyond Rs 1,80,000, ₹12% (before tax).

3. Earnings per share ₹4.

4. Dividend payout ratio is 50% of earnings.

5. Expected growth rate in dividend is 10%.

6. Current market price per share is ₹44

7. Tax rate is 35%.

- You are required to calculate (i) The pattern for raising the additional finance, assuming the firm intends to maintain its existing debt equity mix.

(ii) The after tax average cost of additional debt.

(iii) After tax cost of retained earnings and cost of equity

(iv) WACC as per marginal value weights.

(i)

→ Debt - 3,00,000] 30:70 debt equity mix for 10 lakhs
Equity - 7,00,000

2,10,000 Retained earnings

4,90,000 New equity

(ii) Debt - ₹ 3,00,000

1,80,000 at 6.5% (10 - 35% - 6.5)

1,20,000 at 7.8% (~~10~~ 12 - 35% = 7.8)

2 costs involved hai therefore weighted:

Amount	W		COC		W.C
1,80,000	0.60	×	0.065	=	0.0390
1,20,000	0.40	×	0.078	=	0.0312
3,00,000	1.00				0.0702

$K_d = 7.02\%$

Answer = 7.02%

(iii) $P_0 = ₹ 44$
$g = 10\%$

Earning per share (₹4)

(given) 50% | 50%

Dividend ₹2 | Retained ₹2.

$$D_1 = D_0 (1+g)$$
$$= 2(1+0.10)$$
$$= 2.2$$

$$K_e = \frac{D_1}{P_0} + g$$
$$= \frac{2.2}{44} + 0.10$$
$$= 0.05 + 0.10$$
$$= 0.15$$
$$= 15\%$$

$K_{RE} = K_e$

$\therefore K_{RE} = 15\%$

$\therefore K_e = K_{RE} = 15\%$

(iv)

	Amount	W	COC	W.C
Debt	3,00,000	0.30	0.0702	0.02106
Equity	7,00,000	0.70	0.1500	0.1050
	10,00,000	1.00		0.12606

$\therefore = 12.61\%$

Q) As a financial analyst of a large electronics company, required to calculate WACC as per book value weig[hts] and market value weights.

Source of capital		Amount (₹)
Debentures (₹100 each)	8000 units	8,00,000 (Book
Preference shares (₹100 each)		2,00,000
Equity shares (₹10 each)		10,00,000
Retained earnings		8,00,000
		28,00,000.

All these securities are traded in the capital market recent prices are:

(i) Debentures at ₹ 110 each
(ii) Preference shares at ₹ 120 each
(iii) Equity share at ₹ 22.5 each

Anticipated external investing opportunities are as follow[s]

(i) ₹100 debenture redeemable at par, 10 years maturity, coupon rate, 4% floatation cost, sell price ₹10[0]

(ii) ₹100 preference shares redeemable at par, 10 years [maturity], 12% dividend rate, 5% floatation cost, sell price

(iii) Equity shares, 0.5 ₹ per share floatation cost, sell price ₹ 22.5 each.

In addition the dividend expected on the equity share at the end of the year is ₹ 2.2 per share, the anticipated growth rate is 7% on such dividend. Corporate tax rate is 35%. Cost of retained earnings is 15%.

→ Finding the market value first,

then K_d, K_p, K_e nikalna hai. K_{RE} diya hua hai.

NP in K_d = 96 (100 - 4%)

NP in K_p = 95 (100 - 5%)

~~N~~ P_0 in K_e = 22 (22.5 - 0.5 floatation cost)

K_{RE} = 15% (given)

Q) From the following info, calculate appropriate weighted average cost of capital.

Source	Amount (₹)
Equity shares (1,00,000)	38,00,000
Preference shares	8,00,000
Debentures	50,00,000
Bank Loan (LT)	18,00,000

X	Bank Loan (ST)	both not considered	14,00,000
X	Trade creditors		6,00,000

Additional info:

1. Equity shares include the existing 60,000 shares having current market value of ₹40 per share and the balance is net proceeds from the new issue in the current year (issue price of share is ₹40, floatation cost per share is ₹5). The projected EPS and DPS for the current year are ₹8 and ₹5 respectively.

2. Dividend indicated on preference shares is 12%.

3. Pre tax cost of debenture ₹ 11%.

4. Interest on bank loan is 12% (LT), ~~११~~ 11.5% (ST).

5. Corporate tax rate is 35%, dividend tax is 10%.

6. Market value of preference share is 8,50,000.

→ Market value ke hisaab se WACC find karna hai.

• [Trade creditors bhi short term finance hai, isliye not to be considered]

38,00,000

Existing / \ New

60,000 × 40 = 24,00,000

40,000 ∴ 14,00,000.

$\therefore$ NP for new $= \frac{14,00,000}{40,000}$

$= \underline{\underline{35}}$.

New aur existing dono ka K_e alag aayega.

$\therefore P_0$ in existing $= 40$

in new $= 35$

* $D_1 = D_0 (1 + \underset{\substack{\downarrow \\ 10\% \text{ (given)}}}{gdt})$

$$K_p = \frac{PD(1+dt) + \left(\frac{RV-NP}{n}\right)}{\left(\frac{RV+NP}{2}\right)}$$